THE HEISENBERG UNCERTAINTY PRINCIPLE

LEARNING OBJECTIVES:

By the end of this lecture, you will ability to:

- **Use the 2 versions of Heisenberg's uncertainty principle in calculations.**
- **Explain the implications of Heisenberg's uncertainty principle for measurements.**

Heisenberg's uncertainty principle shows one of the most significant ways of differentiating between classical physics and quantum mechanics

The wave-particle dual nature of matter is the major cause of the origin of the uncertainty principle.
The physical origin of
Heisenberg's uncertainty principle
is with the quantum system

Heisenberg's uncertainty principle states that it is impossible to *simultaneously* calculate or measure with perfect accuracy both the position and the momentum of an object.

Heisenberg's uncertainty principle can be ignored in the macroscopic world (the uncertainties in the position and velocity of objects with relatively large masses are negligible).

It holds significant value in the quantum world. Since atoms and subatomic particles have very small masses, any increase in the accuracy of their positions will be accompanied by an increase in the uncertainty associated with their velocities or momenta.

Determination of position by performing a measurement on the system disturbs it sufficiently to make the determination of momentum imprecise and vice-versa

Heisenberg's uncertainty principle is applicable to all matter waves.

The measurement error of any two conjugate properties, whose dimensions happen to be joule sec, like position - momentum, the time - energy will be guided by the Heisenberg's principle.

But, it will be noticeable and of significance only for small particles like an electron with very small mass.

A bigger particle with a heavy mass will show the error to be very small and negligible.

HEISENBERG UNCERTAINTY PRINCIPLE RELATIONS:

Heisenberg's uncertainty principle can be seen as a very precise mathematical statement that describes the nature of quantum systems.

$$\Delta X \cdot \Delta p \sim \hbar \quad (1)$$

$$\Delta E \cdot \Delta t \sim \hbar \quad (2)$$

Where,

$\hbar$ = value of Planck's constant divided by 2π

ΔX = uncertainty in the position

Δp = uncertainty in momentum

ΔE = uncertainty in the energy

Δt = uncertainty in time measurement

Heisenberg uncertainty principle.

According to Heisenberg's work on quantum mechanics and wave characteristics of particles; for simultaneously measuring the position and momentum of an electron (or any particle), there is an uncertainty in position Δx that is approximately equal to the wavelength λ of the particle.

$$\Delta x \approx \lambda \qquad (3)$$

A wave is not located at one point in space. If the electron's position were to be measured repeatedly, a spread in locations will be observed, implying an uncertainty in position Δx.

To detect the position of the particle, it must be interacted with, such as having it collide with a detector.

In the collision, the particle will lose momentum. This change in momentum could be anywhere from close to zero to the total momentum of the particle, $p = h / \lambda$

It is impossible to say how much momentum will be transferred to a detector, and so there is an uncertainty in momentum Δp**.**

The uncertainty in momentum may be as large as the momentum itself, i. e.

$$\Delta p \approx \frac{h}{\lambda} \qquad (4)$$

The uncertainty in position can be reduced by using a shorter-wavelength electron, since $\Delta x \approx \lambda$.
But shortening the wavelength increases the uncertainty in momentum, since $\Delta p \approx h / \lambda$.
Conversely, the uncertainty in momentum can be reduced by using a longer-wavelength electron, but this increases the uncertainty in position.
Mathematically, this trade-off can be expressed by multiplying the two uncertainties.
The wavelength cancels, leaving

$$\Delta x \Delta p \approx h \qquad \textbf{(5)}$$

So if one uncertainty is reduced, the other must increase so that their product is $\approx h$

In a simultaneous measurement of position and momentum, Heisenberg showed that the best that can be done is

$$\Delta x \Delta p \geq \frac{h}{4\pi} \qquad (6)$$

It is impossible to measure position x and momentum p simultaneously with uncertainties Δx and Δp that multiply to be less than $h/4\pi$.
Neither uncertainty can be zero.
Neither uncertainty can become small without the other becoming large.

A small wavelength allows accurate position measurement, but it increases the momentum of the probe to the point that it further disturbs the momentum of a system being measured

Heisenberg Uncertainty for Energy and Time

Another form of Heisenberg's uncertainty principle for simultaneous measurements of energy and time is

$$\Delta E \Delta t \geq \frac{h}{4\pi} \qquad (7)$$

where ΔE is the uncertainty in energy and Δt is the uncertainty in time.

This means that within a time interval Δt , it is not possible to measure energy precisely - there will be an uncertainty ΔE in the measurement.

In order to measure energy more precisely (to make ΔE smaller), we must increase Δt.
This time interval may be the amount of time used to make the measurement,
or it could be the amount of time a particular state exists.

The uncertainty principle for energy and time can be of great significance if the lifetime of a system is very short.
Then Δt is very small, and ΔE is consequently very large.

A few nuclei and exotic particles have extremely short lifetimes (as small as $10^{-25} s$), causing uncertainties in energy as great as many GeV ($10^{9} eV$).
Stored energy appears as increased rest mass, and so this means that there is significant uncertainty in the rest mass of short-lived particles.
When measured repeatedly, a spread of masses or decay energies are obtained.
The spread is ΔE .

Solved Example on Heisenberg's Uncertainty Principle:

Q. **If the position of the electron is measured within an accuracy of + 0.002 nm, calculate the uncertainty in the momentum of the electron.**

Suppose the momentum of the electron is (h/4pm) × 0.05 nm.

Is there any problem in defining this value?

Solution

a.

$$\Delta x = +0.002nm = 2 \times 10^{-12}m$$

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

$$\Delta p \geq \frac{h}{4\pi} . \left(\frac{1}{\Delta x} \right)$$

$$\Delta p \geq \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 2 \times 10^{-12}}$$

$$\Delta p \geq 2.64 \times 10^{-23} kgms^{-1}$$

b. Momentum p:

$$p = \left(\frac{h}{4pm}\right)(0.05nm) = 8.28 \times 10^{-33} kgms^{-1}$$

$$\frac{\Delta p}{p} = \frac{2.64 \times 10^{-23}}{8.28 \times 10^{-33}} = 3.2 \times 10^{9}$$

Comparing Δp to p; Δp is 3.2×10^{9} larger than p.
Hence there is a problem in defining the given momentum value

ASSIGNMENTS

1. (a) If the position of an electron in an atom is measured to an accuracy of 0.0100 nm, what is the electron's uncertainty in velocity?
   (b) If the electron has this velocity, what is its kinetic energy in eV?

2. The position of a chloride ion on a material can be determined to a maximum error of 1μm. If the mass of the chloride ion is 5.86×10^{-26}kg, what will be the error in its velocity measurement?

3. The lifetime of an excited state of an atom is 3×10^{-3}s. What is the minimum uncertainty in its energy in eV?

4. Determine the minimum uncertainties in the positions of the electron if their speeds are known with a precision of 3.0×10^{-3}m/s